

An introduction to distributionally robust optimization

M. Merino  B. Urrutia 

University of the Basque Country, UPV/EHU maria.merino@ehu.eus

University of the Basque Country, UPV/EHU b.urrutia0621@gmail.com

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Abstract

Initially developed in 1937, Operational Research focused on constructing deterministic models in order to describe and analyze real-world problems to aid decision-making. It was not until the early 1950s when uncertainty was incorporated into these models. Traditionally, two primary disciplines have been employed: Stochastic Optimization (SO) and Robust Optimization (RO). In SO, uncertain parameters are modeled as random variables with known probability distributions. However, there has been significant criticism of its optimistic results in recent years. In contrast, RO addresses uncertainty by considering the worst-case scenarios, leading to over-conservative decisions for other more likely scenarios. An emerging paradigm Distributionally Robust Optimization (DRO) has recently garnered significant attention due to its potential to address the limitations of both SO and RO. DRO serves as a unifying framework by introducing the concept of an ambiguity set, which encompasses a family of distributions deemed *close-enough* to a reference distribution.

Keywords: Distributionally Robust Optimization, Emergency Medical Services, Facility Location, Robust Optimization, Stochastic Optimization.

1 Introduction

Operational Research (OR) is the area of mathematics that mainly assists in decision-making by generating analytic models. One of the widest known techniques in this field is optimization (mathematical optimization or mathematical programming), which maximizes or minimizes one or more objective functions via mathematical modeling of a given problem.

The origin of OR can be traced back to World War II when logistic and strategic decisions focused on military problems were needed. However, it quickly spread to a wide range of

contexts. At the end of the 1940s, George Dantzig proposed a pioneering algorithm to solve the so-called Deterministic Linear Optimization problem (Dantzig, 1948). Uncertainty over the parameters of the problem was first considered in the early 1950s and a new area, Stochastic Optimization (SO), emerged. Dantzig and Madansky were one of the first mathematicians to work in that field (Dantzig & Madansky, 1961). A need for more conservative decisions led the researchers to introduce the so-called Robust Optimization (RO), which instead of optimizing the expectation of a random variable as SO did, aimed to obtain the best results in the worst possible cases.

The beginnings of the min-max problem are attributed to Wald (1949), whose author was also a pioneer in considering two-stage problems as the result of his work on statistical decision functions. In spite of this, RO would not be considered an independent discipline until the work on “Robustness and Optimality as Criteria for Strategic Decisions” (Rosenhead, Elton, & Gupta, 1972).

One of the drawbacks of SO is the need for the distributions of random variables, which are commonly unknown. In the 2010s, a modern and advanced version of SO and RO called Distributionally Robust Optimization (DRO), which considered the so-called ambiguity sets, was presented. These are sets of distributions within a given distance from a reference distribution. That can be measured by the Wasserstein distance. The properties of that distance and the related optimal transportation problem are explained in Villani (2009) by the winner of the Fields Medal in 2010 for his work in the field of Landau damping and the Boltzmann equation.

This paper aims to introduce SO, RO, and DRO frameworks. These will be illustrated by means of a Facility Location Problem for Emergency Medical Services (FLP-EMS). The problem involves locating ambulances within a specific territory to minimize response time to emergency calls and location costs, subject to certain constraints.

Efficient optimization solvers are now already developed that allow linear optimization problems to be easily solved; however, SO, RO, and DRO problems must be reformulated in order to use that software. We will say that a tractable reformulation is achieved when a software is able to solve the problem in an acceptable time. SO and RO have been tractably reformulated in numerous ways, but DRO is an emerging field and its reformulations have recently begun to be developed. An exact reformulation was published in Mohajerin Esfahani & Kuhn (2018), and it was adapted to a two-stage facility location problem in Saif & Delage (2021). However, a more intuitive reformulation for the general problem has been recently proposed by Laureano F. Escudero, Antonio Alonso-Ayuso, Juan F. Monge, Domingo Morales, and Leandro Pardo (Alonso-Ayuso, Escudero, Monge, Morales, & Pardo, 2023). These concepts were deeply studied and adapted to FLP-EMS in the master’s thesis by Urrutia (2024).

Section 2 introduces the FLP-EMS and the deterministic model is proposed. Then, the classic Stochastic and Robust approaches are introduced and formulated in Section 3, and their drawbacks are also highlighted. Section 4 aims to introduce DRO, analyze briefly this framework theoretically and formulate it for FLP-EMS. Finally, Section 5 concludes.

2 Facility Location Problem for Emergency Medical Services

Let us suppose that a Health Emergency Department contacted us to help to decide on how the ambulances should be distributed in a territory. We have then considered a set of potential stations where the ambulances can be located and historical data about where the ambulances were needed in a past period of time. Therefore the problem has the following components:

- Objective: to minimize the time that it takes for a set of emergency calls to be attended, as well as the cost of locating each ambulance in the stations.
- Decisions: location of the ambulances and their allocation to the calls.
- Constraints: the ambulances must be located in a station, and every call must be covered.

A first approach to this problem could be to consider data from a representative day and obtain the optimum location for that day. That problem belongs to **Deterministic Optimization**, which can be formulated as follows:

$$z_{DO} = \min_{x \in \mathbb{X}} f(x), \quad (11)$$

where $\mathbb{X}$ is the feasible set (the set of all possible solutions) and f is the objective function.

2.1 Sets, parameters and variables

We define the following sets, parameters, and decision variables in order to write the FLP-EMS problem in terms of a deterministic optimization problem. Let $J = \{1, \dots, m\}$ be the set of potential stations for ambulances. We want to locate the fleet of ambulances in those stations, so the related decision variable will be x_j which is the number of ambulances located in each station $j \in J$. A cost q_j is considered for each located ambulance in the station $j \in J$ in order to the number of ambulances not to increase in an uncontrolled manner. Let us suppose that in a certain period of time, κ calls are made, which must be covered by the ambulances, ideally within the threshold time t_0 .

A grid over the territory where the cells are in the set $I = \{1, \dots, n\}$ will be used to describe the location of the stations and the emergency calls. ξ_i is the number of calls that occurred in cell $i \in I$. Note that the narrower the grid is, the more accurate the model will be, an example can be seen in Figure 1.

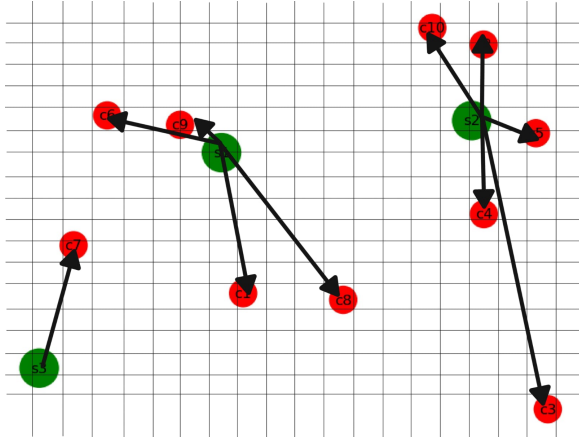


Fig. 1. Example of FLP-EMS problem for $m = 3$ and $\kappa = 10$.

A call is considered to be properly covered if the ambulance arrives at the place in no more than t_0 units of time from when the emergency call is received at the emergency center. Therefore, we will need the time that takes from each station $j \in J$ to each demand point (cell) $i \in I$, which will be denoted by $t_{i,j}$. We will define an integer variable $y_{i,j}$ which describes the number of ambulances dispatched from station j to the cell i . We will introduce the delay, that is, the difference between the response time and the threshold time. The objective function will consider the total delay. Let $c_{i,j}$ be the delay time from the station j to the call i defined as follows

$$c_{i,j} = \max\{0, t_{i,j} - t_0\}. \quad (12)$$

2.2 Deterministic formulation

Once all the sets, parameters, and variables are defined, the deterministic problem can be formulated as follows:

$$\min_{\mathbf{x}, \mathbf{y}} \quad \sum_{j \in J} q_j \cdot x_j + \sum_{i \in I} \sum_{j \in J} c_{i,j} \cdot y_{i,j}, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i \in I} y_{i,j} \leq x_j, \quad \forall j \in J, \quad (13b)$$

$$\sum_{j \in J} y_{i,j} \geq \xi_i, \quad \forall i \in I, \quad (13c)$$

$$x_j \in \mathbb{Z}^+, \quad \forall j \in J, \quad (13d)$$

$$y_{i,j} \in \mathbb{Z}^+, \quad \forall i \in I, \forall j \in J. \quad (13e)$$

Note that (13a) is the sum of the cost of locating the ambulances

and the allocation cost. Moreover, (13b) ensures that no ambulance is allocated from a station if it is not previously located there, and (13c) ensures that every emergency call is covered. (13d) and (13e) come from the definition of the decision variables, but (13e) can be relaxed and considered continuous variables.

One could note that this approach can actually be very far from reality because the variability of the historical data is not taken into consideration and the situation could vary greatly day to day. This will overfit that specific scenario and, therefore, underfit a larger time period. Thus, another mathematical framework is needed to take uncertainty into account.

3 Two classic frameworks under uncertainty

Emergency calls are not deterministic in a more realistic situation. There are two main classic optimization frameworks that take uncertainty into account: Stochastic Optimization and Robust Optimization. Each of those frameworks models the uncertainty in a different way, and they model different objectives as well.

In order to define those models, let us consider the random vector ξ as the one that models the uncertainty, which will take values on a sample space Ω with a distribution $\mathbb{Q}$. Moreover, we will consider discrete random variables, and it can be discretized in continuous cases by means of partitioning the sample space. Let us denote $\mathbb{Q}(\omega)$ the probability of $\omega \in \Omega$ under the distribution $\mathbb{Q}$. In our case, uncertainty will be modeled by using historical data, which will be deemed to involve a set of representative scenarios. These scenarios will be in the sample set $\hat{\Omega}_N = \{\omega_1, \dots, \omega_N\}$, where N is the sample size.

3.1 Stochastic Optimization

We can model uncertainty in the parameters via random variables or random vectors ξ . Each possible realization represents one scenario, that is, ξ^ω denotes the number of calls each cell in the scenario $\omega \in \Omega$, and ξ_i^ω the number of calls in scenario $\omega \in \Omega$ in the cell $i \in I$. The demand parameters can be estimated via historical data within empirical distribution $\hat{\mathbb{P}}_N$ computed, for example, by giving the same probability weight

to each observation:

$$\hat{\mathbb{P}}_N := \frac{1}{N} \sum_{\omega \in \hat{\Omega}_N} \delta_{\xi^\omega}, \quad (14)$$

where δ is the Dirac delta function. Stochastic Optimization classically considers the objective of minimizing the expected value (denoted by $\mathbb{E}$) of the second-stage problem over the sample set with the empirical probability distribution, and the general formulation is as follows:

$$z_{SO} = \min_{x \in \mathbb{X}} \mathbb{E}^{\hat{\mathbb{P}}_N} \left[\min_{y \in \mathbb{Y}(x, \xi)} f(x, \xi, y) \right]. \quad (15)$$

Let us formulate the stochastic model for the FLP-EMS:

$$\min_{\mathbf{x}} \sum_{j \in J} q_j \cdot x_j + \mathbb{E}^{\hat{\mathbb{P}}_N} [g(\mathbf{x}, \omega)], \quad (16a)$$

$$\text{s.t.} \quad \sum_{i \in I} \xi_i^\omega \leq \sum_{j \in J} x_j, \quad \forall \omega \in \hat{\Omega}_N, \quad (16b)$$

$$x_j \in \mathbb{Z}^+, \quad \forall j \in J, \quad (16c)$$

where the function $g(\mathbf{x}, \omega)$, which represents the second-stage problem, is defined as

$$\min_{\mathbf{y}} \sum_{i \in I} \sum_{j \in J} c_{i,j} \cdot y_{i,j}^\omega, \quad (17a)$$

$$\text{s.t.} \quad \sum_{i \in I} y_{i,j}^\omega \leq x_j, \quad \forall j \in J, \quad (17b)$$

$$\sum_{j \in J} y_{i,j}^\omega \geq \xi_i^\omega, \quad \forall i \in I, \quad (17c)$$

$$y_{i,j}^\omega \in \mathbb{Z}^+, \quad \forall i \in I, \forall j \in J. \quad (17d)$$

An additional constraint (16b) was added to the first-stage problem to ensure that the first-stage decision $\mathbf{x}$ remains feasible for every second-stage problem, that is, for every demand realization.

Note that we use the super index ω to denote the dependency of the second-stage variables $\mathbf{y}^\omega$ and the uncertainty realization ξ^ω on the uncertainty. This would be equivalent to solving the allocation problem for each scenario, and minimizing the expectation of the total delay over the set of possible scenarios, assuming that we know the true distribution of probability.

However, this is not a simple linear program because we have an optimization problem within another one. Fortunately, tractable reformulations exist that model stochastic problems as linear programs, see Escudero & Merino (2016), among others.

3.2 Adaptive Robust Optimization

The importance of an emergency call being attended on time is paramount, and just improving the average cases can be an understatement. A more appropriate objective can be to focus on the worst cases. This is what Robust Optimization handles, as it minimizes the worst scenarios. In fact, what RO does is not mitigate the harm that the worst scenario causes, but it makes a decision so that the greatest harm regarding every scenario is as reduced as possible.

In a generic formulation, if our objective is to minimize a cost function, the worst-case would be the maximum of that cost function. Therefore, the maximum is taken with respect to the uncertainty and the master problem then minimizes the harm this situation could cause by making a decision. Hence, the model is as follows

$$z_{RO} = \min_{x \in \mathbb{X}} \max_{u \in \mathcal{U}} f(x, u), \quad (18)$$

where $\mathcal{U}$ is the set with all the possible uncertainty realizations and $f(x, u)$ is the objective function that depends on the decision $x \in \mathbb{X}$ and on the uncertainty realization $u \in \mathcal{U}$. This set $\mathcal{U}$ is also known as the robust set, and it is usually denoted Ω .

As a two-stage problem is being considered in FLP-EMS, a more accurate formulation would be the so-called Adaptive Robust Optimization, which allows a second-stage decision to be made once the uncertainty is fixed:

$$z_{ARO} = \min_{x \in \mathbb{X}} \max_{u \in \mathcal{U}} \min_{y \in \mathbb{Y}(x, u)} f(x, u, y). \quad (19)$$

The ARO formulation for the FLP-EMS is as follows:

$$\min_{\mathbf{x}} \left[\sum_{j \in J} q_j \cdot x_j + \max_{\omega \in \hat{\Omega}_N} \left[\min_{\mathbf{y}^\omega} \sum_{i \in I} \sum_{j \in J} c_{i,j} \cdot y_{i,j}^\omega \right] \right], \quad (20a)$$

$$\text{s.t.} \quad \sum_{i \in I} \xi_i^\omega \leq \sum_{j \in J} x_j, \quad \forall \omega \in \hat{\Omega}_N, \quad (20b)$$

$$\sum_{i \in I} y_{i,j}^\omega \leq x_j, \quad \forall j \in J, \forall \omega \in \hat{\Omega}_N, \quad (20c)$$

$$\sum_{j \in J} y_{i,j}^\omega \geq \xi_i^\omega, \quad \forall i \in I, \forall \omega \in \hat{\Omega}_N, \quad (20d)$$

$$x_j \in \mathbb{Z}^+, \quad \forall j \in J, \quad (20e)$$

$$y_{i,j}^\omega \in \mathbb{Z}^+, \quad \forall i \in I, \forall j \in J, \forall \omega \in \hat{\Omega}_N. \quad (20f)$$

Note that the formulation in (20) is not the one of a single-stage linear program and, therefore, a tractable reformulation is needed.

3.3 Objections

Several objections have been raised regarding both SO and RO. For instance, a reduced amount of historical data can cause the result being over-fitted to that training data and this can lead to under-performances when such results are implemented.

According to SO, this framework returns optimistic results which could lead to post-decision disappointment when implemented, see Mohajerin Esfahani & Kuhn (2018). Moreover, as the objective is to optimize the expected value of certain optimization problems, the variance of the randomness is not taken into account. This can lead to underperformance with variables with large variance, see Hu, Chen, & He (2021). This can result in catastrophic results for specific scenarios.

Since RO focuses on optimizing decisions based on the worst-case scenarios, it can cause less favorable results than those of SO; therefore, this decision may not be the best for other more likely scenarios, see Roos & den Hertog (2020), among others.

4 Distributionally Robust Optimization

This section aims to explain the mathematical basis of Distributionally Robust Optimization models. Moreover, three desirable properties are stated in the joint work Bertsimas, Gupta, & Kallus (2018). These properties are the ones that make a data-driven model strong and solid. These are:

1. **Finite-sample guarantee:** since having infinite data is not realistic, we want our method to be close to reality even with limited data. We will consider that we have N observations.
2. **Asymptotic performance guarantee:** we want to ensure that if we have an infinite amount of data, our model will be precise. That is, the larger the sample size is, the better our model is.
3. **Tractability:** since we need to obtain solutions, the problem cannot be hard from the computational point of view.

In Mohajerin Esfahani & Kuhn (2018) those properties are proven under the next assumption:

Assumption 4.1 (Light-tailed distribution). *If we call $\mathbb{Q}$ the real distribution of the random vector ξ , there exists a parameter $a > 1$ (which we will call growth rate) such that*

$$\mathbb{E}^{\mathbb{Q}}[\exp(\|\xi\|^a)] = \int_{\Omega} \exp(\|\xi^{\omega}\|^a) \mathbb{Q}(d\omega) < \infty. \quad (21)$$

The tractability part just consists of reformulating the model in a linear convex program (Mohajerin Esfahani & Kuhn, 2018).

4.1 DRO general model

The keys to propose an adequate RO model are the way to model the objective function and the way to build the robust set, see Conejo (2019a) and Conejo (2019b). Let us consider the following formulation:

$$\min_{x \in \mathbb{X}} \max_{u \in \mathcal{U}} \min_{y \in \mathbb{Y}(x, u, \xi)} \mathbb{E}^{\mathbb{Q}}[f(x, u, y, \xi)], \quad (22)$$

where ξ is the random vector modeling uncertainty. This is called **Adaptive Robust Stochastic Optimization (ARSO)**, which is a hybrid between RO and SO. Instead of optimizing the worst-case (RO) or the expectation (SO) of the second stage problem, ARSO optimizes the worst-case expectation. When combining robust optimization with stochastic programming thus, the apprehension will be taken into consideration in two different ways, which can be helpful in modeling both long-term and short-term uncertainties. In general, robustness would be desirable in the long-term, because these decisions are usually not easily modifiable; and stochastic in the short-term because short-term decisions are usually adjustable.

Concerning the robust set, instead of using an approximated distribution, optimizing over the possible distributions the parameters can follow based on historical data is usually done. The only difference is that the robust set is now composed of distributions. Hence, the formulation is as follows:

$$z_{DRO} = \min_{x \in \mathbb{X}} \max_{\mathbb{P} \in B_{\varepsilon}(\hat{\mathbb{P}}_N)} \min_{y \in \mathbb{Y}(x, \mathbb{P}, \xi)} \mathbb{E}^{\mathbb{P}}[f(x, y, \xi)], \quad (23)$$

where ξ follows the distribution $\mathbb{P}$, $\hat{\mathbb{P}}_N$ is an empirical probability distribution obtained from the historical data and $B_{\varepsilon}(\hat{\mathbb{P}}_N)$ is a set of distributions close to the empirical distribution which is properly detailed in Definition 4.2. This is precisely **Distributionally Robust Optimization (DRO)** framework. Note that as the distribution does not depend on the second-stage decision, and assuming certain conditions over f such as linearity, this would be equivalent to

$$\min_{x \in \mathbb{X}} \max_{\mathbb{P} \in B_{\varepsilon}(\hat{\mathbb{P}}_N)} \mathbb{E}^{\mathbb{P}} \left[\min_{y \in \mathbb{Y}(x, \mathbb{P}, \xi)} f(x, y, \xi) \right], \quad (24)$$

which leads us to the so-called *general formulation* for DRO:

$$\min_{x \in \mathbb{X}} \max_{\mathbb{P} \in B_{\varepsilon}(\hat{\mathbb{P}}_N)} \mathbb{E}^{\mathbb{P}}[g(x, \xi)]. \quad (25)$$

4.2 Wasserstein distance

A distance between distributions needs to be defined in order to define what “close enough distributions” means. One of the most relevant metrics for probability measures is the Wasserstein metric, although this is not the only metric defined for probability measures. For example, the *Lévi-Prokhorov metric* (Prokhorov, 1956), the *total variation distance* (Parzen, 1960) or other integral probability metrics can be found in the literature. The preference for the Wasserstein metric is supported by two main arguments. Firstly, the Wasserstein distance is defined as an optimization problem of minimization, making it easier to find upper bounds for the distance and define where the limit for close enough is. Secondly, the distance can be dualized which makes the reformulation of the problems easier. In Villani (2009) a deeper comparison between distances is made.

We will denote by $\mathcal{M}(\Omega)$ all the probability distributions $\mathbb{P}$ supported on a sample space Ω such that $\mathbb{E}^{\mathbb{P}}[\|\xi\|] < \infty$, where ξ is a random vector whose components are defined in a probability space Ω . Note that as the norm is a measurable function, $\|\xi\|$ is another random variable. Then, the Wasserstein distance is defined as follows:

Definition 4.1. Let $\Omega \subseteq \mathbb{R}^n$ and $p \geq 1$ be a real number. We denote by $\Pi(\mathbb{P}, \mathbb{Q})$ the set of the probabilities in $\Omega \times \Omega$ with marginal distributions $\mathbb{P}$ and $\mathbb{Q}$. Then, the p -Wasserstein distance or the p -Wasserstein metric is a distance in $\mathcal{M}(\Omega)$ defined as:

$$W_p: \mathcal{M}(\Omega) \times \mathcal{M}(\Omega) \longrightarrow \mathbb{R}^+, \\ (\mathbb{P}, \mathbb{Q}) \longmapsto \inf_{\pi \in \Pi(\mathbb{P}, \mathbb{Q})} \{ \mathbb{E}^{\pi} [\|\xi - \zeta\|_p] \},$$

where $\|\cdot\|_p$ is the p -norm in $\mathbb{R}^n$, and ξ and ζ are the random vectors that the distributions $\mathbb{P}$ and $\mathbb{Q}$ define, respectively. When $p = 1$ we will just call it Wasserstein distance.

Note that this is just the optimal transportation plan, where what we transport is the probability mass of one distribution to another. As we are considering discrete random variables, the problem is as follows:

$$W_p(\mathbb{P}, \mathbb{Q}) = \min_{\pi \geq 0} \sum_{i=1}^N \sum_{j=1}^N \|\xi_i - \zeta_j\|_p \pi_{i,j}, \quad (26a)$$

$$\text{s.t. } \sum_{j=1}^N \pi_{i,j} = \mathbb{P}_i, \quad \forall i = 1, \dots, N, \quad (26b)$$

$$\sum_{i=1}^N \pi_{i,j} = \mathbb{Q}_j, \quad \forall j = 1, \dots, N. \quad (26c)$$

Let us illustrate an example of Wasserstein distance between two distributions $\mathbb{P}$ and $\mathbb{Q}$. Let the sample set be $\Omega = \{\omega_1, \dots, \omega_{10}\} = \{1, \dots, 10\}$ and the following two probability distributions

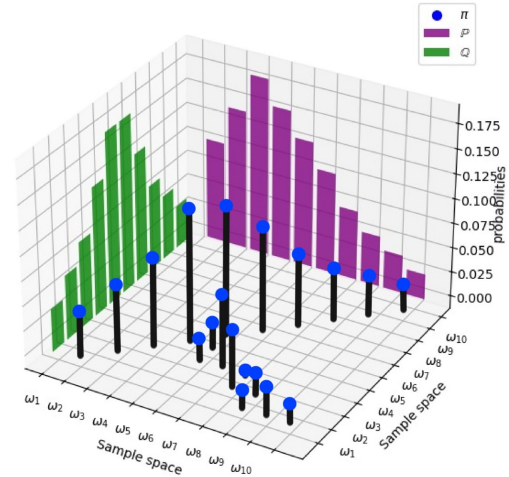


Fig. 2. Example of an optimal transportation plan for the Wasserstein distance.

$$\mathbb{P} = (0.10, 0.14, 0.18, 0.16, 0.13, 0.10, 0.07, 0.05, 0.04, 0.03) \quad (27)$$

and

$$\mathbb{Q} = (0.04, 0.07, 0.09, 0.13, 0.17, 0.17, 0.13, 0.09, 0.07, 0.04) \quad (28)$$

over the sample set Ω , colored in purple and in green, respectively, in Figure 2. The optimal transportation plan is represented by the black columns, where the height represents the amount of probability mass transported from $\mathbb{P}$ to $\mathbb{Q}$ or in the symmetric way. In this case, the 1-Wasserstein distance is 1.14 units.

Remember that our aim is to build a robust set, which is composed by probability measures close to the real measure. We can build the ambiguity set now that we have a distance to measure the closeness.

Definition 4.2. Let $\varepsilon > 0$ be a real number, $p \in [1, +\infty)$ and $\mathbb{P}$ a distribution defined in some Ω . Then, we will call the Wasserstein ball of order p , radius ε and with center $\mathbb{P}$ the set

$$B_\varepsilon^p(\mathbb{P}) := \{ \mathbb{Q} \in \mathcal{M}(\Omega) : W_p(\mathbb{P}, \mathbb{Q}) \leq \varepsilon \}.$$

When talking in general terms or when p is already specified, we will simplify the notation by using just $B_\varepsilon(\mathbb{P})$.

4.3 Example on FLP-EMS

In order to optimize over the Wasserstein ball of radius ε , a set of probability distributions within distance ε from reference distribution will be considered. In Figure 3, we illustrate three distributions over the set $\Omega_3 = \{\omega_1, \omega_2, \omega_3\}$ for FLP-EMS. Let us call $\mathbb{P}_1$, $\mathbb{P}_2$ and $\mathbb{P}_3$ the distributions of the random vector relative to the set of calls $\xi = (\xi^\omega)_{\omega \in \Omega}$, where $1/3$ of probability is assigned to each scenario in the sample set and 0 otherwise. The grid considered is of size 20×20 . We will compute the distance from $\mathbb{P}_1$ to the other two, where the first distribution is semi-colored.

The results (in units of distance) of the computed Wasserstein distances are as follows:

$$\begin{aligned} W(\mathbb{P}_1, \mathbb{P}_1) &= 0, \\ W(\mathbb{P}_1, \mathbb{P}_2) &= \frac{14}{3} = 4.67, \\ W(\mathbb{P}_1, \mathbb{P}_3) &= \frac{268}{15} = 17.87. \end{aligned} \quad (29)$$

As expected, the distance gets larger when the scenarios in the sample are further from each other.

4.4 DRO properties

The finite sample guarantee seeks an upper bound of the worst-case expectation, which is obtained under the real distribution $\mathbb{Q}$. Therefore, if we call $\hat{z}_N$ to the optimum value obtained with the data set, what we want is a solution $\widehat{x}_N$ such that

$$\mathbb{E}^{\mathbb{Q}}[g(\widehat{x}_N, \xi)] \leq \hat{z}_N. \quad (30)$$

If this holds it would mean that the optimal value does not underestimate the reality, and therefore, this would guarantee the robustness of the solution. In order to obtain that, we consider the following theorem:

Theorem 4.1 (Measure Concentration Theorem, Mohajerin Esfahani & Kuhn, 2018). *If Assumption 4.1 holds, for all $N \geq 1$, $n \neq 2$ and $\varepsilon > 0$, there exist some constants $c_1 > 0$ and $c_2 > 0$ (that only depend on m , on the exponential growth rate and on the expected value of the exponential of the variable), such that*

$$\mathbb{Q}^N \left\{ W(\mathbb{Q}, \widehat{\mathbb{P}}_N) \geq \varepsilon \right\} \leq \begin{cases} c_1 \cdot \exp(-c_2 \cdot N \cdot \varepsilon^{\max\{n, 2\}}), & \text{if } \varepsilon \leq 1, \\ c_1 \cdot \exp(-c_2 \cdot N \cdot \varepsilon^a), & \text{if } \varepsilon > 1. \end{cases} \quad (31)$$

It would be desirable to consider the radius of the Wasserstein ball such that the true distribution is inside the Wasserstein ball under a confidence level $1 - \alpha$ with $\alpha \in (0, 1)$ as follows:

$$\mathbb{Q}^N \left\{ W(\mathbb{Q}, \widehat{\mathbb{P}}_N) \geq \varepsilon \right\} \leq \alpha \quad (32)$$

This would completely determine the radius $\varepsilon_N(\alpha)$ if those constants c_1 and c_2 were known. However, The Measure Concentration Theorem (4.1) leads us to the finite sample guarantee, as was our aim.

Theorem 4.2 (Finite Sample Guarantee, Mohajerin Esfahani & Kuhn, 2018). *If Assumption 4.1 holds, then the worst-case expectation is bounded by $\hat{z}_N$ with probability $1 - \alpha$.*

It is trivial to see that for a fixed α , the radius $\varepsilon_N(\alpha)$ converges to 0 as the size of the sample N grows. That means that the ambiguity set will converge to the center of the Wasserstein ball. Consequently, we can ensure with a probability $1 - \alpha$ that our empirical distribution coincides with the true distribution when N tends to infinity. This implies that the distributionally robust model will converge to the stochastic model with perfect information for large sample sizes.

Nonetheless, fixing a value for α is very restrictive, which could lead us to consider varying the value of the probability so that this converges to 0; therefore the statement above holds with probability 1 when the sample size N tends to infinity. This is the intuitive idea of the asymptotic performance guarantee, ensuring that the solution $\widehat{x}_N$ and the certificate $\hat{z}_N$ converge (with probability 1) to the solution and the optimal value of the stochastic problem, respectively.

To prove that, the next lemma will be helpful, especially with the part corresponding to the distribution. Then, the theorem in which this lemma is used states the part corresponding to the solution and the optimum value.

Lemma 4.3 (Convergence of distributions, Mohajerin Esfahani & Kuhn, 2018). *Let us suppose that the true distribution is light-tailed (Assumption 4.1). Let $(\alpha_N)_{N \in \mathbb{N}}$ be a sequence of parameters where:*

$$(a) \alpha_N \in (0, 1), \forall N \in \mathbb{N}, (b) \sum_{N=1}^{\infty} \alpha_N < \infty, (c) \lim_{N \rightarrow \infty} \varepsilon_N(\alpha_N) = 0$$

Then, every distribution $\widehat{\mathbb{Q}}_N \in B_{\varepsilon_N(\alpha_N)}(\widehat{\mathbb{P}}_N)$ (where $N \in \mathbb{N}$) almost surely converges with respect to the Wasserstein metric

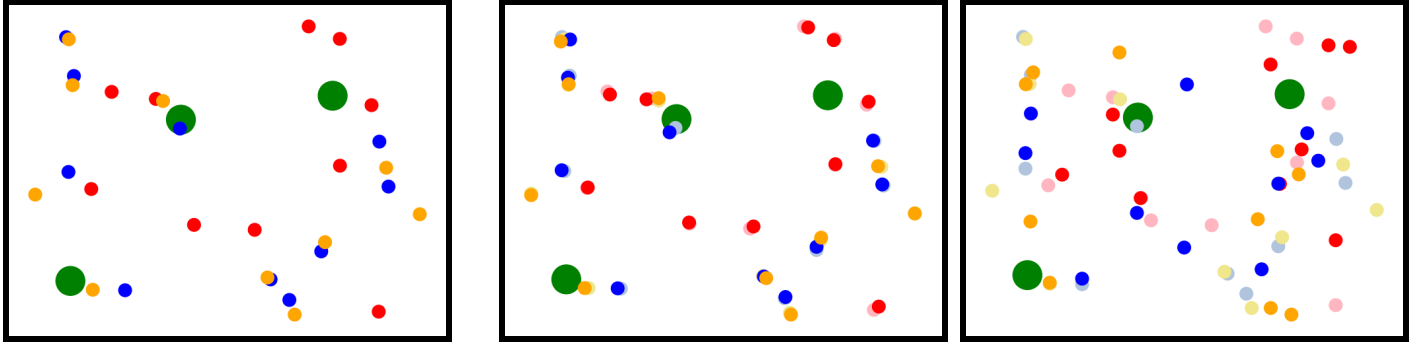


Fig. 3. Example of three distributions $\mathbb{P}_1$ (left), $\mathbb{P}_2$ (center) and $\mathbb{P}_3$ (right) with 3 scenarios (in yellow, red and blue)

to the true distribution $\mathbb{Q}$, that is,

$$\mathbb{Q}^\infty \left\{ \lim_{N \rightarrow \infty} W(\mathbb{Q}, \hat{\mathbb{Q}}_N) = 0 \right\} = 1. \quad (33)$$

Theorem 4.4 (Asymptotic Performance Guarantee, Mohajerin Esfahani & Kuhn, 2018). *Let us suppose that the true distribution is light-tailed (Assumption 4.1). Let $(\alpha_N)_{N \in \mathbb{N}}$ be a sequence of parameters where: (a) $\alpha_N \in (0, 1)$, $\forall N \in \mathbb{N}$, (b) $\sum_{N=1}^{\infty} \alpha_N < \infty$, (c) $\lim_{N \rightarrow \infty} \varepsilon_N(\alpha_N) = 0$*

Then, if the loss function $g(x, \xi)$ is upper semicontinuous in ξ and if there exists a constant $L \geq 0$ such that

$$|g(x, \xi)| \leq L \cdot (1 + \|\xi\|), \quad \forall x \in \mathbb{X},$$

1. **Convergence of the optimum value:** *the certificate $\hat{z}_N$ almost surely converges to the real optimum value z^* when the sample size tends to infinity.*
2. **Convergence of the optimum solution:** *and if $\mathbb{X}$ is closed and the loss function $g(x, \xi^\omega)$ is lower semicontinuous in x for every $\omega \in \Omega$, then every accumulation point of $\{\hat{x}_N\}_{N \in \mathbb{N}}$ is almost surely an optimal solution.*

4.5 DRO formulation for FLP-EMS

Once the theoretical strength of this framework has been underlined, the model is as follows:

$$\min_{\mathbf{x}} \sum_{j \in J} q_j \cdot x_j + \max_{\mathbb{P} \in \mathcal{B}_{\varepsilon_N}(\hat{\mathbb{P}}_N)} \mathbb{E}^{\mathbb{P}} [g(\mathbf{x}, \omega)] \quad (34a)$$

$$\text{s.t.} \quad \sum_{i \in I} \xi_i^\omega \leq \sum_{j \in J} x_j \quad \forall \omega \in \hat{\Omega}_N \quad (34b)$$

$$x_j \in \mathbb{Z}^+ \quad \forall j \in J \quad (34c)$$

where g is defined as in (17).

5 Comparison and conclusions

The reformulation in SO and ARO frameworks have been widely studied. In contrast, DRO is an emerging framework that has gained attention recently due to the robustness of its solutions when applied to out-of-sample data.

With respect to the theoretical results, it is interesting to remark that DRO meets the objective to make a hybridization between SO and ARO. Firstly, we ensure (Urrutia, 2024) that this framework can be very efficient when applied out of sample even with very small data, which is not the case with SO. Asymptotic performance guarantee ensures that the more data we have the better the solution is; this is a desirable property when data-based problems are being solved.

In practice, the three frameworks can be implemented since tractable reformulations exist. Despite that, recent works such as Mohajerin Esfahani & Kuhn (2018) and Saif & Delage (2021) among others, experimentally showed that the results of DRO are promising even if the computational expense is high.

For further details, the tractable reformulations of the presented (16), (20) and (34) models are detailed in the masters thesis Urrutia (2024). Moreover, it is shown experimentally that

as the Wasserstein radius gets smaller, DRO converges to SO, and that as the Wasserstein radius increases, DRO takes more conservative decisions converging to RO. Moreover, experimentation also showed that post-decision disappointment decreases as the radius increases.

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About the authors



María Merino Maestre is an associate professor in the department of Mathematics at the University of the Basque Country. Her main research interests include Stochastic Programming and Combinatorial Optimization. She focuses on designing mathematical models that incorporate uncertainty and develop risk management strategies. She is working on algorithms to solve large-scale optimization problems, with the aim of achieving a balance between quality, efficiency and speedup.



Beñat Urrutia Redondo holds a Degree in Mathematics from the University of the Basque Country and completed the inter-university Master's program in Mathematical Modeling and Research, Statistics, and Computing. Beñat also enrolled in a five-month internship at the BCAM -Basque Center for Applied Mathematics-, where he analyzed Distributionally Robust Optimization in the context of an Emergency Medical Services problem.

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